

Haokai Hong

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Biography

Haokai Hong is currently a Ph.D. student in the Department of Data Science and Artificial Intelligence at The Hong Kong Polytechnic University (PolyU), Hong Kong SAR, China, under the supervision of Prof. Kay Chen Tan, Professor at The Hong Kong Polytechnic University. Prior to joining PolyU, he received the Bachelor degree and Master degree from the School of Informatics, Xiamen University, Xiamen, China, in 2020 and 2023, respectively. His research interests include artificial intelligence for science, molecular dynamics simulation, geometric deep learning, generative modeling, and large-scale multiobjective optimization.

Education

The Hong Kong Polytechnic University (PolyU)	Hong Kong SAR, China
Ph.D. in Data Science and Artificial Intelligence	2023 – Present
Supervisor: Prof. Kay Chen Tan	
Xiamen University	Xiamen, China
Master of Engineering, School of Informatics	2020 – 2023
Supervisor: Prof. Min Jiang	
Xiamen University	Xiamen, China
Bachelor Degree, School of Informatics	2016 – 2020

Publications

Conference Papers

1. **H. Hong**, W. Lin, and K. C. Tan. “Large-Scale Molecular Dynamics Simulations: Direct Interatomic Modeling with Dilated Message Passing.” In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.
2. **H. Hong**, W. Lin, C. Zhang, and K. C. Tan. “Geometric Graph Neural Diffusion for Stable Molecular Dynamics Simulations.” In *The 14th International Conference on Learning Representations (ICLR)*, 2026.
3. **H. Hong**, W. Lin, M. Yang, and K. C. Tan. “Distributional Priors Guided Diffusion for Generating 3D Molecules in Low Data Regimes.” In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, Oral, 2026.
4. **H. Hong**, W. Lin, and K. C. Tan. “Accelerating 3D Molecule Generation via Jointly Geometric Optimal Transport.” In *The 13th International Conference on Learning Representations (ICLR)*, 2025.
5. **H. Hong**, M. Jiang, L. Feng, Q. Lin, and K. C. Tan. “Balancing Exploration and Exploitation for Solving Large-Scale Multiobjective Optimization via Attention Mechanism.” In *Proceedings of the IEEE Congress on Evolutionary Computation (CEC)*, pp. 1–8, 2022.

6. **H. Hong**, K. Ye, M. Jiang, and K. C. Tan. “Solving Large-Scale Multi-Objective Optimization via Probabilistic Prediction Model.” In *Evolutionary Multi-Criterion Optimization (EMO)*, LNCS 12654, Springer, 2021.

Journal Papers

7. **H. Hong**, M. Jiang, and G. G. Yen. “Boosting Scalability for Large-Scale Multiobjective Optimization via Transfer Weights.” *Information Sciences*, vol. 670, 2024.
8. **H. Hong**, M. Jiang, Q. Lin, and K. C. Tan. “Efficiently Tackling Million-Dimensional Multiobjective Problems: A Direction Sampling and Fine-Tuning Approach.” *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2024.
9. **H. Hong**, M. Jiang, and G. G. Yen. “Improving Performance Insensitivity of Large-Scale Multiobjective Optimization via Monte Carlo Tree Search.” *IEEE Transactions on Cybernetics*, 2023.
10. **H. Hong**, K. Ye, M. Jiang, D. Cao, and K. C. Tan. “Solving Large-Scale Multiobjective Optimization via the Probabilistic Prediction Model.” *Memetic Computing*, 14(2):165–177, 2022.

Selected Collaborations

1. C. Chen, **H. Hong***, W. Lin, and K. C. Tan. “A Physics-Informed Evolutionary Transfer Optimization Framework for Material Design.” In *Proceedings of the IEEE Congress on Evolutionary Computation (CEC)*, pp. 1–9, 2025.
2. L. Cao, **H. Hong**, and M. Jiang. “Fast Solving Partial Differential Equations via Imitative Fourier Neural Operator.” In *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*, pp. 1–8, 2024.
3. M. Jiang, Z. Wang, **H. Hong**, and G. G. Yen. “Knee Point-Based Imbalanced Transfer Learning for Dynamic Multiobjective Optimization.” *IEEE Transactions on Evolutionary Computation*, 25(1):117–129, 2021.
4. Z. Wang, **H. Hong**, K. Ye, G.-E. Zhang, M. Jiang, and K. C. Tan. “Manifold Interpolation for Large-Scale Multiobjective Optimization via Generative Adversarial Networks.” *IEEE Transactions on Neural Networks and Learning Systems*, 2021.

Teaching Assistance

2024 – 2025

- COMP 6707: Advanced Computational Intelligence
- COMP 5511: Artificial Intelligence Concepts

2023 – 2024

- COMP 6707: Advanced Computational Intelligence
- COMP 5511: Artificial Intelligence Concepts

Academic Services

- Reviewer for international conferences and journals in artificial intelligence and computational intelligence, including ICML, NeurIPS, TEVC, and TNNLS.